DATA SCIENCE AND VISUALIZATION

Project Planning and Dataset Selection

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team Number: Team 9

1. INTRODUCTION

The first week focused on team formation, dataset selection, defining the problem statement and objectives, and planning the project workflow. The team selected a Spotify tracks dataset containing information about songs, artists, albums, popularity, and audio features.

The team also learned about Pandas and NumPy for data analysis and identified the tools required for the project. These activities established the foundation for exploring the relationship between musical characteristics and song popularity in the upcoming stages.

2. PROJECT TITLE

Discovering the Factors Behind Song Popularity Through Data Science

3. PROBLEM STATEMENT

Music streaming platforms contain a large number of songs with different musical characteristics and popularity levels. Understanding the characteristics associated with song popularity can help reveal meaningful patterns in music data.

This project aims to analyze Spotify track data using data science and visualization techniques. The project will involve understanding the dataset, assessing its quality, handling missing values and inconsistencies, performing exploratory data analysis, and investigating relationships between audio features and popularity.

The findings will be communicated through suitable visualizations and, in the later stages, an interactive dashboard.

4. OBJECTIVES

The main objectives of the project are:

- To select and understand the Spotify dataset and its attributes.
- To learn Pandas and NumPy for data analysis.
- To identify data-quality issues and perform exploratory data analysis.
- To analyze relationships between musical features and popularity, create visualizations, and prepare data for advanced analysis and dashboard development.

5. TEAM FORMATION

The project team consists of four members who will collaboratively work on dataset preparation, analysis, visualization, documentation, and presentation.

S. No.	Team Member	Register Number
1	Prithesa R	25BAD083
2	Saravanan A	25BAD102
3	Tejas T	25BAD124
4	Viknesh T	25BAD128

The team will share responsibilities to ensure that each stage of the project is completed systematically. Specific individual task assignments can be documented once finalized by the team.

6. DATASET SELECTION

6.1 Dataset Description

The team selected the Spotify tracks dataset, containing 19,335 records and 25 columns covering song, artist, album, popularity, and audio characteristics.

6.2 Important Attributes

The dataset contains the following categories of information:

- **Track information:** Track ID, track name, track number, and album details.
- **Artist information:** Artist names associated with tracks.
- **Album information:** Album name, album type, release date, and number of tracks in an album.

- **Popularity information:** Popularity values associated with tracks.
- **Audio features:** Danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.
- **Musical attributes:** Key, mode, and time signature.

These attributes provide a combination of numerical, categorical, and textual data that can be examined using statistical and visualization techniques.

6.3 Reasons for Dataset Selection

The Spotify dataset was selected for the following reasons:

1. Represents real-world music data with details about songs, artists, and popularity.
2. Contains multiple attributes and audio features suitable for statistical and comparative analysis.
3. Provides opportunities to practise data cleaning, preprocessing, exploratory data analysis, correlation analysis, and visualization.
4. Supports advanced techniques such as feature engineering and Principal Component Analysis (PCA).

7. INTRODUCTION TO PANDAS

7.1 Definition

Pandas is an open-source Python library used for data manipulation and analysis. It provides two important data structures: Series and DataFrame.

A Series represents one-dimensional labelled data, whereas a DataFrame represents two-dimensional tabular data arranged in rows and columns.

7.2 Applications of Pandas

Pandas provides several useful operations for working with datasets:

1. Loading datasets from CSV and other file formats and displaying the first and last few records.
2. Examining the number of rows and columns, column names, and data types.
3. Identifying missing values and duplicate records and generating descriptive statistics.
4. Filtering, sorting, and transforming data for further analysis.

7.3 Application in Our Project

Pandas will be used to load and inspect the Spotify dataset. It will help the team understand the dataset structure, identify data-quality issues, and prepare the data for cleaning and exploratory analysis.

Example:

```
import pandas as pd
df = pd.read_csv("DSV_TEAM-9_RAW_DATASET.csv")
print(df.head())
print(df.shape)
```

The read_csv() function loads the dataset, head() displays the first five rows by default, shape returns the number of rows and columns.

8. INTRODUCTION TO NUMPY

8.1 Definition

NumPy, which stands for Numerical Python, is a Python library used for numerical computing. It provides the multidimensional array structure and mathematical functions required for efficient numerical operations.

8.2 Applications of NumPy

NumPy supports:

- Creating and manipulating numerical arrays for data processing.
- Performing mathematical and statistical calculations, including mean, median, and standard deviation.
- Carrying out element-wise operations and numerical transformations.
- Supporting data preprocessing, statistical analysis, and future PCA workflows.

8.3 Application in Our Project

NumPy will be used to perform numerical calculations on relevant Spotify audio features. It can help calculate statistical measures and prepare numerical data for further analysis.

Example:

```
import numpy as np
audio_features = np.array([0.75, 0.82, 0.68, 0.90, 0.73])
```

```
print("Average Audio Feature:", np.mean(audio_features))
print("Standard Deviation:", np.std(audio_features))
```

The `array()` function creates a numerical array, `mean()` calculates its average, and `std()` calculates its standard deviation.

9. PROPOSED PROJECT METHODOLOGY

The project will follow a systematic data science workflow to study Spotify song popularity and explore relationships between musical characteristics and popularity.

The planned workflow is:

1. **Raw Dataset Selection:** Select the Spotify tracks dataset for the project.
2. **Data Understanding:** Study the dataset structure, attributes, and meaning of the available information.
3. **Data Cleaning:** Plan the preparation of the dataset for reliable analysis.
4. **Exploratory Data Analysis:** Explore patterns and relationships among relevant variables.
5. **Feature Engineering:** Prepare and transform suitable attributes for analysis.
6. **Feature Selection:** Identify relevant features for investigating song popularity.
7. **Principal Component Analysis (PCA):** Explore dimensionality reduction where appropriate.
8. **Data Visualization:** Develop charts and visual representations of the findings.
9. **Insights and Data Storytelling:** Present the major findings in an understandable form.

12. TOOLS AND TECHNOLOGIES

The following tools and technologies are planned for the project:

- **Python:** Primary programming language for data analysis.
- **Pandas:** Data manipulation and handling tabular datasets.
- **NumPy:** Numerical calculations and mathematical operations.
- **Matplotlib:** Creating graphs and charts.
- **Seaborn:** Statistical data visualization.
- **Scikit-learn:** Supporting feature processing and PCA.
- **Jupyter Notebook:** Writing, executing, and documenting Python code.
- **GitHub:** Version control and team collaboration.
- **Streamlit or Plotly:** Potential tools for developing an interactive dashboard.

These tools will support the different stages of the project, from data preparation to visualization and presentation.

13. WEEK 1 ACTIVITIES AND MILESTONES

The main activities planned and undertaken during Week 1 were:

- Formed the project team, selected the Spotify tracks dataset, and finalized the project title.
- Defined the problem statement, project objectives, and overall project scope.
- Introduced Pandas and NumPy and identified the required software tools and libraries.
- Planned the project workflow, established GitHub for collaboration, and prepared the initial documentation.

These activities established the foundation for the subsequent stages of the project.

14. EXPECTED OUTCOMES

The expected outcomes of the project are:

- Understand the Spotify dataset and its attributes.
- Analyze the relationship between audio features and song popularity.
- Create visualizations to identify trends and important features.
- Develop an interactive dashboard to present the findings and insights.

15. CONCLUSION

Week 1 established the foundation for the Data Science and Visualization project titled *Discovering the Factors Behind Song Popularity Through Data Science*.

The team selected the Spotify tracks dataset, defined the problem statement and objectives, and planned the overall project methodology. The team also introduced Pandas and NumPy as essential Python libraries for data manipulation and numerical computation. GitHub was identified as the platform for maintaining project files and supporting collaboration.

These initial activities provide a clear direction for the project. In the following weeks, the team will proceed with the scheduled stages of dataset understanding, preparation, analysis, and visualization to investigate the factors associated with Spotify song popularity.

DATA SCIENCE AND VISUALIZATION-WEEK 2 REPORT

Know Your Data

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team: 9

1. Introduction

Data understanding is an essential stage in the data science workflow. Before performing data cleaning, statistical analysis, or visualization, it is necessary to examine the structure, characteristics, and quality of the dataset.

During Week 2, the Spotify Songs dataset was explored using Python, Pandas, and NumPy. The analysis focused on inspecting the dataset, understanding its data types, calculating preliminary descriptive statistics, identifying missing values and duplicate records, and generating an automated First Impression Data Quality Report.

2. Aim

To load, inspect, and understand the structure and quality of the Spotify Songs dataset using Python, Pandas, and NumPy.

3. Objectives

1. Load the Spotify Songs dataset from a CSV file using Pandas.
2. Examine the dataset dimensions, first and last five records, column names, and data types.
3. Calculate preliminary descriptive statistics for numerical and categorical attributes.
4. Identify missing values, duplicate records, and potential data quality issues.
5. Generate an automated First Impression Data Quality Report and export the results as CSV files.

4. Tools and Technologies Used

- **Python:** Used for dataset exploration and analysis.
- **Pandas:** Used to load, inspect, and summarize the dataset.
- **NumPy:** Used for numerical operations.
- **Jupyter Notebook:** Used to execute the code and document the results.
- **CSV:** Used for storing the original dataset and exported reports.

5. Dataset Overview

The Spotify Songs dataset was obtained from Kaggle and contains information about songs, artists, albums, popularity, and audio characteristics.

The dataset includes attributes such as track name, artist information, album details, popularity, danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.

Dataset details:

Attribute	Description
Total records	19,335
Total columns	46
File format	CSV
Data types	String, float, and integer
Memory usage	Approximately 74.65 MB

Each row represents a song record, while the columns contain information describing its metadata and audio features.

6. Dataset Loading and Inspection

The dataset was loaded into a Pandas DataFrame using `pd.read_csv()`. The `head()` and `tail()` functions were used to display the first five and last five records.

The `shape`, `columns`, `dtypes`, and `info()` functions were used to examine the dataset structure, column names, data types, non-null counts, and memory usage.

Observation: The dataset contains 19,335 rows and 46 columns. It includes both numerical audio features and textual metadata. Some attributes that represent numerical information are stored as strings and may require further validation.

7. Preliminary Descriptive Statistics

The `describe()` function was used to obtain an initial statistical summary of numerical and categorical attributes.

For numerical columns, the summary included count, mean, standard deviation, minimum, quartiles, and maximum. For categorical columns, it included the number of non-null entries, unique values, the most frequent value, and its frequency.

The numerical summary provided an initial understanding of the audio features. For example, the mean danceability was approximately **0.616**, mean energy was **0.584**, and mean tempo was approximately **119.79 BPM**.

Observation: The summaries provided an overview of the dataset's values and distributions. Detailed statistical analysis was left for a later stage of the project.

8. Missing Value Analysis

Missing values were identified using the `isnull()` and `sum()` functions. Missing percentages were calculated to evaluate the completeness of individual columns.

The analysis produced the following results:

Metric	Result
Total missing cells	7,707
Rows containing missing values	7,532
Columns containing missing values	39
Missing values in `preview_url`	7,524
Missing percentage in `preview_url`	38.91%

The `preview_url` column contained the largest number of missing values. Several audio feature and metadata columns also contained missing values, although some had only seven missing entries each.

Observation: Missing values were identified and documented without modifying the dataset. Appropriate handling decisions were reserved for the data cleaning stage.

9. Duplicate Record Analysis

The `duplicated()` function was used to identify exact duplicate rows. The Spotify song identifier column, `id`, was also checked separately for repeated identifiers.

The results were:

- **Exact duplicate rows:** 0
- **Repeated Spotify IDs:** 2

Observation: No exact duplicate rows were found. However, two repeated Spotify IDs were identified and require further investigation before deciding whether any records should be removed.

No records were deleted during Week 2.

10. Data Type Analysis

The data types of all columns were examined using `dtypes` and a data type summary table.

The dataset contained:

Data type	Number of columns
String (`str`)	32
Float (`float64`)	13
Integer (`int64`)	1
Total	**46**

The analysis also checked whether selected fields containing numerical information could be converted into numeric values.

Column	Invalid numerical values
`popularity`	1
`track_number`	1
`duration_ms.x`	4

Observation: Some numerical fields were stored as strings, and a small number of values could not be converted directly into numbers. These inconsistencies were documented for further examination. No data type corrections were made during Week 2.

11. Automated First Impression Data Quality Report

An automated First Impression Data Quality Report was generated using Pandas. It summarized the condition of all 46 columns and included:

- Column names and data types.
- Non-null counts.
- Missing value counts and percentages.
- Number of unique values.

A separate dataset summary was also generated to record the total rows, total columns, missing cells, rows containing missing values, columns containing missing values, exact duplicate rows, and memory usage.

The following CSV reports were exported successfully:

1. `Week2_Data_Quality_Report.csv`
2. `Week2_Dataset_Summary.csv`

Observation: The reports provide a structured overview of the dataset's completeness and quality. They will serve as references for planning the next stage of preprocessing.

12. Team Member Participation

The following is a proposed allocation of responsibilities based on the activities performed during Week 2.

S. No.	Team Member	Roll Number	Contribution
1	Prithesa R	25BAD083	Dataset loading, initial inspection, and understanding the dataset structure.
2	Saravanan A	25BAD102	Preliminary descriptive statistics and data type analysis.
3	Tejas T	25BAD124	Missing value analysis and duplicate record verification.
4	Viknesh T	25BAD128	Automated data quality report generation and report documentation.

13. Challenges and Considerations

1. The dataset contains missing values across multiple columns, particularly `preview_url`.
2. Some numerical attributes are stored as strings and require validation.
3. Repeated Spotify IDs need investigation even though no exact duplicate rows were found.
4. Metadata and URL fields require consideration when preparing the dataset for further analysis.

These findings were documented without changing the original dataset.

14. Conclusion

During Week 2, the Spotify Songs dataset was successfully loaded and examined using Python, Pandas, and NumPy. Its structure, data types, preliminary descriptive statistics, missing values, and duplicate records were analyzed.

The investigation identified 7,707 missing cells, no exact duplicate rows, two repeated Spotify IDs, and several values that require numerical validation. An automated First Impression Data Quality Report and a dataset summary were generated and exported successfully.

The original dataset was preserved without modification. The findings establish a foundation for the data cleaning and preprocessing activities in Week 3.

DATA SCIENCE AND VISUALIZATION - WEEK 3 REPORT

Spotify Dataset Cleaning ion and Outlier Detect

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team 9

Dataset: Spotify Tracks Dataset

1. Introduction

Week 3 focused on cleaning and preparing the Spotify dataset for further analysis. Data cleaning is an essential step in data science because missing values, duplicate records, and extreme observations can affect the reliability of analytical results.

In this stage, Python was used to identify and handle missing values, check duplicate records, and detect potential outliers using statistical methods and boxplots. The cleaned dataset was then verified and prepared for subsequent analysis.

2. Objectives

The main objectives of Week 3 were:

1. Identify missing values and handle them using suitable methods.
2. Check the dataset for duplicate records.
3. Detect potential outliers in numerical audio features.
4. Compare outlier detection results using statistical methods.
5. Verify and save the cleaned dataset for further analysis.

3. Tools and Technologies Used

The following tools and libraries were used:

- **Python:** Programming language for data cleaning and analysis.
- **Pandas:** Used to inspect, clean, summarize, and export the dataset.

- **NumPy:** Used for numerical operations.
- **Matplotlib:** Used to display graphs and visualizations.
- **Seaborn:** Used to generate boxplots.
- **Jupyter Notebook:** Used to execute Python code and display results.

4. Dataset Overview

The Spotify dataset contains information about tracks, albums, popularity, Spotify identifiers, and audio characteristics. The dataset was loaded and inspected using Pandas.

The notebook reported **19,335 rows and 46 columns**. Nine numerical audio features were selected for outlier detection:

1. Danceability
2. Energy
3. Loudness
4. Speechiness
5. Acousticness
6. Instrumentalness
7. Liveness
8. Valence
9. Tempo

These features were examined to understand their distributions and identify potential extreme observations.

5. Data Cleaning Methodology

5.1 Identification of Missing Values

Missing values were checked across the dataset by calculating the number of missing entries in each column.

The preview_url column contained 7,524 missing values, representing approximately 38.91% of the dataset. Several audio-related fields contained seven missing values each, representing approximately 0.04%.

This assessment helped identify columns that required attention during data cleaning.

5.2 Handling Missing Values

Missing values were handled using suitable imputation methods:

- **Numerical columns:** Missing values were filled using the median of the respective column.
- **Categorical or object columns:** Missing values were filled using the mode where a valid mode was available.
- **Verification:** The dataset was checked again after imputation to identify any remaining missing values.

The final missing-value check reported no remaining missing values.

5.3 Duplicate Record Detection

The dataset was checked for exact duplicate rows using Pandas. The drop_duplicates() method was applied to the working dataset.

The notebook reported zero duplicate rows. Therefore, the dataset remained at 19,335 rows and 46 columns after the duplicate-checking step

6. Outlier Detection and Visualization

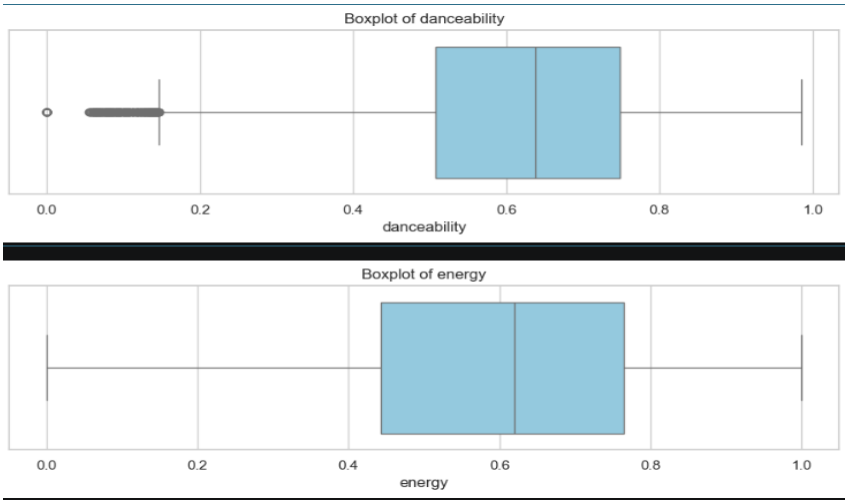
Outlier detection was performed to identify values that differed considerably from the general distribution of the selected audio features. Boxplots and statistical methods were used to identify potential outliers.

6.1 Boxplot Analysis

Boxplots were used to visualize the distributions of the nine numerical audio features. They display the median, interquartile range, and potential extreme observations.

The analysis identified potential outliers in several features, particularly instrumentalness, speechiness, loudness, and liveness. The presence of these values does not necessarily indicate incorrect data because extreme audio-feature values may represent genuine characteristics of songs.





6.2 Interquartile Range (IQR) Method

The IQR method is used to detect outliers by measuring the spread of the middle 50% of the data. It calculates the difference between the third quartile (Q3) and the first quartile (Q1).

IQR = Q3 - Q1

The method defines an acceptable range using the following limits:

- Lower limit: (Q1 - 1.5 times IQR)
- Upper limit: (Q3 + 1.5 \times IQR)

Any observation falling outside this range is identified as a potential outlier. This method was applied to the selected Spotify audio features to identify unusually low or high values.

6.3 Z-Score Method

The Z-score method is used to identify outliers by measuring how far an observed value deviates from the mean in terms of standard deviations.

[Z = frac{x-mu}{sigma}]

Where:

- (x) – Observed value
- (mu) – Mean of the dataset
- (sigma) – Standard deviation

In this study, values with an absolute Z-score greater than 3 ((|Z| > 3)) were identified as potential outliers. This method helped detect unusually high or low values in the selected Spotify audio features.

6.4 Comparison of Outlier Detection Results

The number of potential outliers identified by the IQR and Z-score methods is summarized below.

Audio Feature	IQR Outliers	IQR Percentage	Z-Score Outliers
Danceability	258	1.33%	90
Energy	0	0.00%	0
Loudness	1,813	9.38%	479
Speechiness	2,255	11.66%	336
Acousticness	0	0.00%	0
Instrumentalness	4,274	22.10%	886
Liveness	1,404	7.26%	524
Valence	0	0.00%	0
Tempo	70	0.36%	53

The IQR method identified the highest number of potential outliers in instrumentalness, followed by speechiness and loudness. The Z-score method also identified instrumentalness as the feature with the highest number of potential outliers, followed by liveness and loudness.

The difference in counts occurs because the two methods use different statistical criteria. These results indicate potential outliers rather than confirmed data errors.

6.5 Outlier-Handling Decision

Potential outliers were retained instead of being automatically removed. Extreme values in audio features may represent genuine differences between songs. Removing them without further investigation could result in the loss of useful information.

Therefore, outlier detection was used to identify observations for review rather than automatically delete them.

7. Descriptive Statistical Analysis

Descriptive statistics were examined to understand the central tendency and range of the selected audio features.

Audio Feature	Mean	Minimum	Maximum
Danceability	0.616	0.000	0.985
Energy	0.584	0.00002	1.000
Loudness	-8.605 dB	-54.376 dB	1.060 dB
Speechiness	0.107	0.000	0.954
Acousticness	0.309	0.000	0.996
Instrumentalness	0.102	0.000	1.000
Liveness	0.182	0.000	0.993
Valence	0.491	0.000	1.000
Tempo	119.795 BPM	0.000 BPM	230.401 BPM

The results show variation across the selected audio features. Danceability and energy had mean values of 0.616 and 0.584, respectively. Instrumentalness had a mean of 0.102, while the average tempo was approximately 119.795 BPM.

These statistics provide a numerical summary of the selected features and support further analysis of their relationship with song popularity.

8. Final Dataset Verification

After cleaning and outlier detection, the dataset was checked to confirm its final condition.

Verification	Result
Final number of rows	19,335
Final number of columns	46
Remaining missing values	0

Duplicate rows reported	0
Rows removed solely due to outlier flags	0
Exported file	spotify_cleaned.csv

The final checks indicated that the dataset contained no remaining missing values and no exact duplicate rows. Potential outliers were retained for further interpretation.

The cleaned dataset was exported as spotify_cleaned.csv for use in subsequent stages of the project.

9. Team Member Participation

The tasks for Week 3 were distributed among the team members to ensure systematic data preprocessing, analysis, and documentation.

S. No.	Team Member	Roll Number	Tasks and Responsibilities
1	Prithesa R	25BAD083	Dataset understanding, identification of missing values, and handling missing data using suitable imputation techniques.
2	Saravanan A	25BAD102	Duplicate record verification, data consistency checks, and final dataset validation after cleaning.
3	Tejas T	25BAD124	Outlier detection using the Interquartile Range (IQR) method and visualization of outliers using boxplots.
4	Viknesh T	25BAD128	Z-score-based outlier detection, descriptive statistical analysis, and preparation of the report documenting the results.

10. Challenges and Considerations

The following considerations were identified during the data-cleaning process:

1. The dataset contains metadata and URL fields that need careful handling.
2. The `preview_url` column contains many missing values.
3. IQR and Z-score methods identified different numbers of outliers.
4. Extreme values were retained, as they may represent valid musical characteristics. Further analysis is needed to study their relationship with song popularity.

11. Conclusion

Week 3 focused on cleaning and preparing the Spotify dataset for further analysis. Missing values were handled using median and mode imputation, duplicate records were checked, and potential outliers were investigated using boxplots, the IQR method, and Z-scores.

The final verification reported 19,335 rows and 46 columns, with no remaining missing values and no exact duplicate rows. Potential outliers were retained rather than automatically removed because extreme values may represent genuine musical characteristics.

The cleaned dataset was exported as spotify_cleaned.csv and is prepared for the next stage of the project: exploratory data analysis.

DATA SCIENCE AND VISUALIZATION-WEEK 4 REPORT

Exploratory Data Analysis of Spotify Songs

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team: 9

Dataset: Spotify Songs Dataset

1. Introduction

Week 4 focused on Exploratory Data Analysis (EDA) of the cleaned Spotify Songs dataset to understand song popularity, audio features, and their relationships. EDA helps identify patterns and variations in data using statistical summaries and visualizations.

The analysis was performed using Python in Jupyter Notebook, with Pandas and NumPy for data processing and Matplotlib and Seaborn for visualization. The findings provide initial insights into the audio features associated with song popularity and help guide further analysis.

2. Objectives

The main objectives of Week 4 were:

- To load and inspect the cleaned Spotify Songs dataset, including its structure, columns, and data types.
- To check for missing values and duplicate records in the dataset.
- To calculate descriptive statistics for popularity and selected audio features.
- To analyze the distribution and spread of important numerical variables using visualizations.
  To examine the relationships between song popularity and selected audio features.
- To identify positive and negative correlations and develop preliminary insights into song popularity.

3. Tools and Libraries Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

4. Dataset Overview

The cleaned Spotify Songs dataset was used for Exploratory Data Analysis (EDA) to understand song popularity and audio features.

- Dataset Size: 19,335 rows and 46 columns.
- Features: Includes popularity and audio features such as danceability, energy, loudness, acousticness, valence, and tempo.
- Data Quality: Checked for missing values and duplicate records.
- Data Preprocessing: Converted selected columns into appropriate numerical data types.
- Purpose: To understand the dataset structure and prepare it for statistical analysis and visualization.

5. Descriptive Statistical Analysis

Descriptive statistical analysis was performed to understand the central values and general characteristics of popularity and the selected audio features. The notebook calculated statistical measures such as count, mean, standard deviation, minimum, quartiles, and maximum.

The mean provides the average value, while the median represents the middle observation. Comparing these measures gives an initial indication of the distribution of a variable.

5.1 Summary Statistics

The following table presents selected mean and median values reported in the Week 4 analysis.

Variable	Mean	Median
Popularity	58.14	64.00
Danceability	0.62	0.64
Energy	0.58	0.62

Loudness	-8.60	-6.82
Acousticness	0.31	0.18
Valence	0.49	0.49
Tempo (BPM)	119.79	119.89

5.2 Interpretation of the Statistics

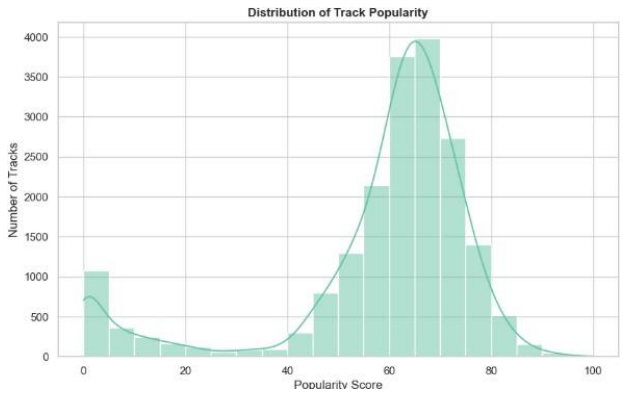
Popularity had a mean of 58.14 and a median of 64.00, suggesting that lower scores may reduce the average. Danceability showed a mean of 0.62 and a median of 0.64, while energy had a mean of 0.58 and a median of 0.62, indicating moderately high levels for both features.

Loudness had a mean of −8.60 dB and a median of −6.82 dB, showing variation across tracks. Acousticness had a mean of 0.31 and a median of 0.18, suggesting some higher values. Valence had a mean and median of 0.49, while tempo had a mean of 119.79 BPM and a median of 119.89 BPM. These statistics provide a baseline for understanding song characteristics and their relationship with popularity.

6. Exploratory Data Analysis and Visualizations

EDA was performed using numerical summaries and visualizations. To keep the report focused, three important graphs are selected for inclusion: the popularity distribution histogram, the audio-features boxplots, and the correlation heatmap. Together, they show the distribution of the target variable, the spread of musical features, and the relationships among numerical variables.

6.1 Distribution of Song Popularity

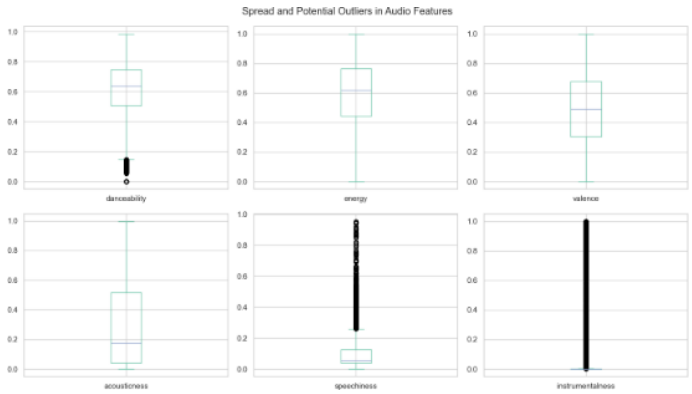


The popularity histogram illustrates the distribution of song popularity scores across the dataset. It groups scores into intervals and shows the number of songs in each interval. This helps

identify where popularity scores are concentrated and whether the distribution is skewed toward lower or higher values.

The mean popularity was 58.14, while the median was 64.00. The lower mean suggests that lower popularity scores may be pulling down the average. Since popularity is the main variable of interest, understanding its distribution provides a useful foundation for analyzing how different musical features relate to song popularity

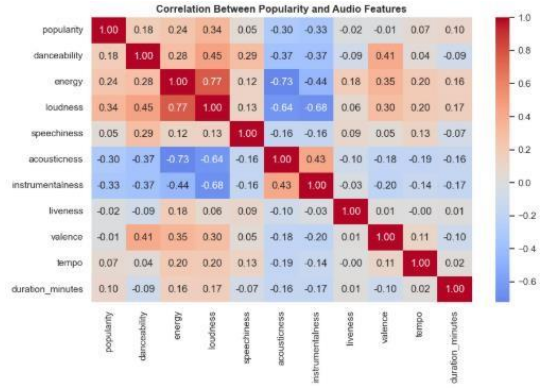
6.2 Spread and Potential Outliers in Audio Features



Boxplots summarize numerical data using the median, quartiles, and whiskers. They help compare the spread of audio features and identify potential outliers. Since features such as danceability, energy, acousticness, and valence range from 0 to 1, while loudness is measured in decibels and tempo in BPM, each feature must be interpreted according to its own scale.

The boxplots show the variation in audio features and highlight extreme values that may represent potential outliers. These values are not necessarily errors, as they may reflect genuine musical characteristics. Overall, this visualization provides a compact comparison of feature distributions and helps identify patterns for further analysis.

6.3 Correlation Between Popularity and Audio Features



A correlation heatmap shows the strength and direction of relationships between numerical variables using a color scale. Values range from -1 to +1: positive correlation means variables tend to increase together, negative correlation means one tends to decrease as the other increases, and values near zero indicate a weak linear relationship. The heatmap helps examine relationships between popularity and audio features, as well as among the features themselves.

In the Week 4 analysis, loudness showed the strongest positive correlation with popularity, while instrumentalness showed the strongest negative correlation. Acousticness also had a negative correlation with popularity. These findings reveal associations between musical features and popularity, but correlation does not prove causation or mean that changing a feature will directly change a song's popularity.

7. Key Findings from the Analysis

The notebook calculated Pearson correlation coefficients to examine the linear relationships between popularity and selected audio features.

7.1 Features Positively Associated with Popularity

Audio feature	Correlation with popularity	Interpretation
Loudness	+0.337	The strongest positive association among the reported features.
Energy	+0.241	A weak positive linear association.
Danceability	+0.178	A weak positive linear association.

1. Loudness: Loudness showed a positive correlation of approximately 0.337 with popularity, indicating that tracks with higher loudness values tended to have higher popularity scores. However, this relationship alone cannot explain popularity.

2. Energy: Energy showed a positive correlation of approximately 0.241, suggesting a weak tendency for tracks with higher energy levels to have higher popularity scores.

3. Danceability: Danceability showed a positive correlation of approximately 0.178, indicating a weak relationship with popularity. This suggests that danceability alone has a limited linear association with song popularity.

7.2 Features Negatively Associated with Popularity

Audio feature	Correlation with popularity	Interpretation
Instrumentalness	-0.330	The strongest negative association among the reported features.
Acousticness	-0.295	A weak-to-moderate negative linear association

Instrumentalness: Instrumentalness had a correlation of approximately -0.330 with popularity. In this dataset, higher instrumentalness values tended to be associated with lower popularity scores.

Acousticness: Acousticness had a correlation of approximately -0.295. This suggests that tracks with higher acousticness values tended to have lower popularity scores within the analyzed data.

These observations describe the patterns found in this dataset and should not be generalized to every song or music category without additional investigation.

7.3 Overall Interpretation

The correlation analysis showed that loudness and energy had positive associations with popularity, while instrumentalness and acousticness had negative associations, and danceability had a weaker positive association. Since these relationships were not strong, song popularity likely depends on multiple factors, including artist recognition, promotion, listener preferences, playlist placement, and release timing. Therefore, audio features are useful indicators but cannot independently explain popularity.

8. Team Member Participation

S. No.	Team Member	Register Number	Contribution
1	Prithesa R	25BAD083	Dataset inspection and exploratory data analysis.

2	Saravanan A	25BAD102	Descriptive statistical analysis and interpretation of results.
3	Tejas T	25BAD124	Visualization of audio features and analysis of their distributions.
4	Viknesh T	25BAD128	Correlation analysis, identification of key findings, and report preparation.

9. Challenges and Considerations

- Missing Values: The popularity column contained one missing value, which should be checked before further analysis.
- Duration Units: The duration variable requires correction because its values were not properly converted from milliseconds to minutes.
- Correlation Interpretation: Correlations indicate relationships between audio features and popularity but do not prove cause and effect.
- Potential Outliers: Extreme values in audio features should be examined carefully, as they may represent genuine musical characteristics rather than errors.
- Multiple Influencing Factors: Song popularity may depend on factors beyond audio features, such as artist recognition, promotion, and listener preferences.

10. Conclusion

Week 4 focused on EDA of the cleaned Spotify dataset using statistical summaries, visualizations, and correlation analysis. Loudness showed the strongest positive correlation with popularity (0.337), while instrumentalness showed the strongest negative correlation (-0.330).

The analysis revealed that multiple audio features are associated with popularity, but no single feature explains it completely. Data-validation issues involving popularity and duration also need attention before proceeding to feature engineering and further analysis.

DATA SCIENCE AND VISUALIZATION-WEEK 6 REPORT

Feature Engineering and Data Preprocessing

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team: 9

Dataset: Spotify Songs Dataset

1. Introduction

Feature engineering and data preprocessing are important stages in preparing data for machine learning. In this week, we worked with Spotify song data to create useful features, handle missing values, transform skewed numerical features, encode categorical variables, and scale numerical data. The main objective was to convert the raw features into a consistent numerical format suitable for future analysis and machine learning models.

2. Objectives

The main objectives of Week 6 were:

- Create new features from existing song information.
- Identify and handle missing values.
- Analyze the distribution and skewness of numerical features.
- Apply the Yeo–Johnson transformation to selected skewed features.
- Encode categorical variables using one-hot encoding.
- Apply standard scaling and min-max normalization.
- Combine the processed features into a machine-learning input dataset while keeping popularity as the target variable.

3. Tools and Libraries Used

- **Python:** Data preprocessing and feature engineering.
- **Pandas and NumPy:** Data manipulation and numerical operations.
- **Matplotlib:** Visualization of feature distributions.

- **Scikit-learn:** Missing-value imputation, power transformation, categorical encoding, and feature scaling.
- **Jupyter Notebook:** Implementation and documentation of the workflow.

4. Dataset Information

The Spotify dataset consists of **1,913 rows and 45 columns**, containing song details, album information, popularity scores, and audio features. These data were used for feature engineering and preprocessing, including missing-value handling, skewness transformation, categorical encoding, and feature scaling. The popularity column was retained as the target variable for further analysis.

5. Feature Engineering

Two new numerical features were created from existing columns:

New Feature	Source Column	Purpose
`duration_minutes`	`duration_ms.x` or `duration_ms.y`	Converts song duration from milliseconds into minutes
`release_year`	`album.release_date`	Extracts the release year from the album release date

These features provide more interpretable information about song duration and release time, making them easier to use in further analysis.

6. Feature Selection and Classification

For preprocessing, the notebook grouped the selected columns into numerical and categorical features.

Feature Type	Number of Original Features	Examples
Numerical	11	Danceability, energy, loudness, tempo, duration, release year
Categorical	5	Explicit, album type, key, mode, time signature
Total	**16**	Selected preprocessing features

The popularity column was kept separate from these input features so that it could serve as the target variable.

7. Missing-Value Handling

Missing values were checked before applying preprocessing techniques. The notebook identified **81 missing values in release_year**. The other selected numerical and categorical columns had no missing values in the displayed check.

Two imputation strategies were used:

- **Median imputation:** Replaced missing numerical values with the median of the respective feature.
- **Most-frequent imputation:** Replaced missing categorical values with the most common category.

After imputation, the notebook reported zero missing values in both the numerical and categorical feature sets.

8. Skewness Analysis and Yeo–Johnson Transformation

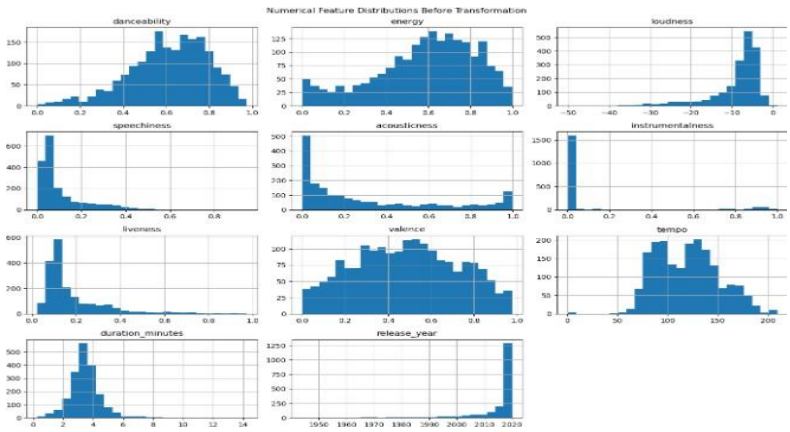
Skewness was calculated to understand whether numerical features were distributed symmetrically or concentrated toward one side. Features with absolute skewness greater than 1 were selected for transformation.

The notebook selected six features: loudness, speechiness, instrumentalness, liveness, duration_minutes, and release_year.

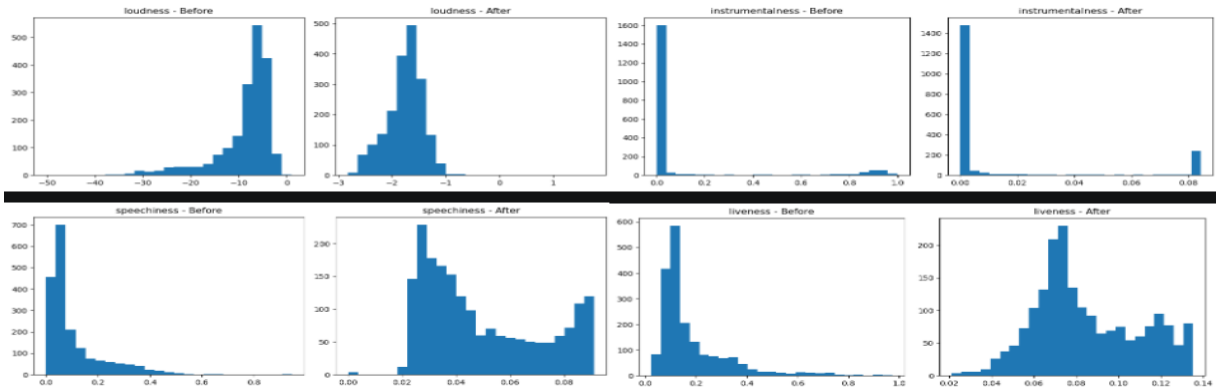
The Yeo–Johnson transformation was applied to reduce skewness. It improved the reported skewness of several features:

Feature	Skewness Before	Skewness After
Loudness	-2.173	0.155
Speechiness	2.350	0.569
Instrumentalness	2.393	1.877
Liveness	2.255	0.343
Duration in minutes	1.818	0.083
Release year	-3.058	Invalid (NaN)

The results show a substantial reduction in skewness for loudness, speechiness, liveness, and duration. Instrumentalness remained positively skewed. The release-year result was affected by numerical overflow and requires correction before the transformation can be considered successful for that feature.



Numerical feature histograms before transformation.



Before-and-after histograms for the selected skewed features

9. Categorical Feature Encoding

Categorical features contain values that represent groups or labels rather than continuous numerical measurements. To prepare these features for further analysis and machine learning, One-Hot Encoding was applied. This method converts each category into a separate binary column containing 0 or 1.

The following five categorical features were selected:

S. No.	Feature Name	Description
1	explicit	Indicates whether a track contains explicit content
2	album.album_type	Represents the type of album release
3	key	Represents the musical key of a track
4	mode	Represents the musical mode
5	time_signature	Represents the time signature of a track

These five categorical features were converted into **24 binary columns**. This makes categorical information suitable for numerical processing while preserving the categories represented by the original features.

10. Feature Scaling and Normalization

10.1 Standard Scaling

StandardScaler was used to standardize the 11 numerical features, bringing them to a comparable scale. It transforms values based on the mean and standard deviation, producing features with a mean of approximately 0 and a standard deviation of approximately 1.

10.2 Min-Max Normalization

MinMaxScaler was used to rescale numerical feature values to a range between 0 and 1. This method helps make features with different value ranges comparable. Standard scaling and min-max normalization were used to prepare the numerical data for further analysis.

11. Combining Processed Features

After numerical and categorical preprocessing, the processed features were combined into a single input feature matrix. This step brings the numerical features and encoded categorical information together in a consistent format.

Component	Number of Features
Numerical features	11
One-hot encoded categorical features	24
Total input features	**35**

The target variable, popularity, was kept separate from the input features because the project aims to study the factors associated with song popularity.

The combined feature matrix provides a structured dataset for further analysis and potential machine-learning applications. The final dimensions and missing-value count should be reported according to the results obtained after executing all processing steps.

12. Team Member Participation

S. No.	Team Member Name	Roll Number	Topics Covered
1	Prithesa R	25BAD083	Feature Selection and Data Cleaning
2	Saravanan A	25BAD102	Categorical Feature Encoding
3	Tejas T	25BAD124	Feature Scaling and Normalization
4	Viknesh T	25BAD128	Feature Engineering and Feature Combination

13. Challenges Faced

The main challenges during feature engineering and preprocessing were:

- Understanding the dataset structure and identifying the relevant numerical and categorical features.
- Handling missing values and analyzing skewness in numerical data.
- Applying Yeo–Johnson transformation to suitable features.
- Choosing appropriate encoding and scaling techniques to prepare the data for further analysis.

14. Conclusion

Week 6 focused on feature engineering and preprocessing of the Spotify dataset. The team created useful features, handled missing values, analyzed skewness, encoded categorical variables, and applied feature scaling and normalization techniques. The processed numerical and categorical features were then combined, while popularity was kept separate as the target variable. These steps prepared the dataset for further analysis and visualization in the upcoming weeks.

DATA SCIENCE AND VISUALIZATION-WEEK 7 REPORT

Feature Selection and Dimensionality Reduction

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team: 9

Dataset: Spotify Songs Dataset

1. Introduction

Week 7 focused on feature selection and dimensionality reduction using the Spotify dataset. As the dataset contains multiple audio features, identifying the most relevant variables and reducing the number of dimensions can simplify analysis while preserving important information. Correlation analysis, SelectKBest, and Principal Component Analysis (PCA) were used to study the relationships between audio features and song popularity.

2. Objectives

The main objectives of Week 7 were:

- Identify audio features associated with song popularity.
- Select the most relevant features using SelectKBest.
- Standardize numerical features before applying PCA.
- Reduce the number of dimensions while retaining most of the data's variance.
- Visualize feature importance, explained variance, and principal components.
- Export the reduced dataset for future analysis.

3. Tools and Libraries Used

- **Python:** Data processing and analysis.
- **Pandas and NumPy:** Data manipulation and numerical calculations.
- **Matplotlib and Seaborn:** Graphs and visualizations.
- **Scikit-learn:** StandardScaler, SelectKBest, f_regression, and PCA.

- **Jupyter Notebook:** Implementation and documentation.

4. Dataset Information

The cleaned Spotify dataset used in this analysis contained **19,335 records and 45 columns**. After handling the missing popularity value, 19,334 records were used.

Nine numerical audio features were selected for analysis:

- Danceability
- Energy
- Loudness
- Speechiness
- Acousticness
- Instrumentalness
- Liveness
- Valence
- Tempo

Popularity was retained as the target variable for correlation and feature-selection analysis, but it was not included among the nine PCA input features. Missing values in the selected audio features were filled using their respective medians.

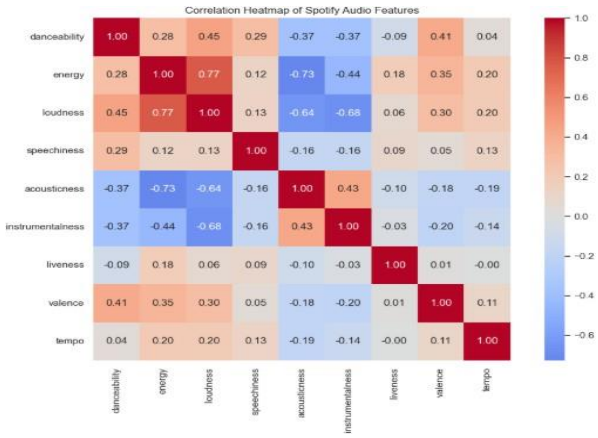
5. Correlation Analysis

Correlation analysis was performed to understand the relationships between individual audio features and popularity. The results showed that loudness had the strongest positive correlation, while instrumentalness and acousticness had the strongest negative correlations.

Audio feature	Correlation with popularity
Loudness	+0.337
Energy	+0.241
Danceability	+0.178
Tempo	+0.067
Speechiness	+0.046
Valence	−0.012
Liveness	−0.016
Acousticness	−0.295

Instrumentalness	−0.330

The results indicate that louder and more energetic tracks tended to have higher popularity scores in this dataset, whereas instrumentalness and acousticness showed negative associations. These correlations describe relationships and do not establish cause and effect.



Audio-feature correlation heatmap

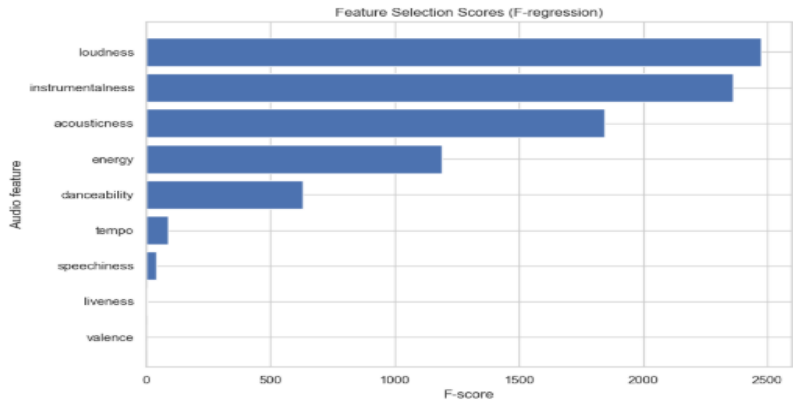
6. Feature Selection Using SelectKBest

The SelectKBest method with the f_regression scoring function was applied to rank the nine audio features according to their statistical relationship with popularity. The five highest-scoring features were selected.

Selected feature	F-score
Loudness	2474.20
Instrumentalness	2359.97
Acousticness	1843.89
Energy	1190.67
Danceability	632.03

Loudness received the highest F-score, followed by instrumentalness and acousticness. Speechiness, liveness, valence, and tempo were not included in the final set of five features.

The F-score measures the strength of the statistical relationship with the target variable; it does not indicate whether the relationship is positive or negative. The feature-selection results provide a smaller set of useful variables for further investigation.



F-score bar chart

7. Standardization of Audio Features

Before applying PCA, the nine numerical audio features were standardized using StandardScaler. Standardization transforms the features to have approximately zero mean and unit standard deviation.

This step is important because audio features such as loudness and tempo use different scales. Without standardization, variables with larger numerical scales could disproportionately influence the principal components.

After standardization, the feature matrix contained **19,334 records and 9 standardized features**.

8. Principal Component Analysis (PCA)

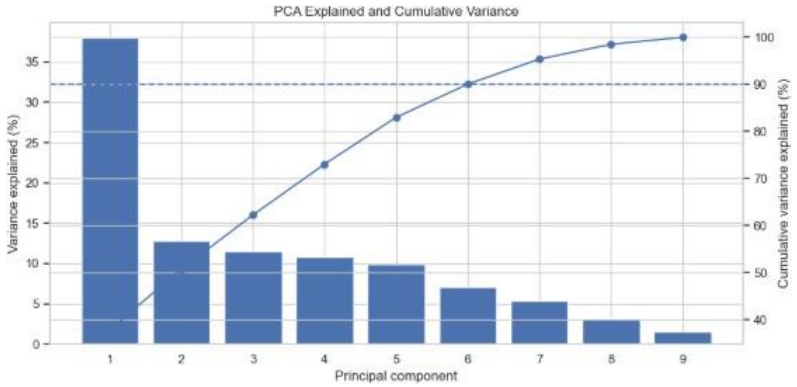
PCA was applied to transform the nine standardized audio features into a smaller set of principal components. Each principal component represents a combination of the original features and captures a portion of the variation in the data.

The cumulative explained variance was used to decide how many components to retain.

Principal component	Explained variance	Cumulative variance
PC1	37.99%	37.99%
PC2	12.81%	50.80%
PC3	11.50%	62.30%
PC4	10.79%	73.09%
PC5	9.94%	83.02%
PC6	Approximately 7.04%	90.06%

The first principal component explained the largest proportion of variance, approximately 37.99%. The first five components explained 83.02%, while six components retained approximately **90.06% of the total variance**.

Therefore, six components were selected to reduce the nine original audio features while preserving most of their variation.



Variance and Cumulative variance chart

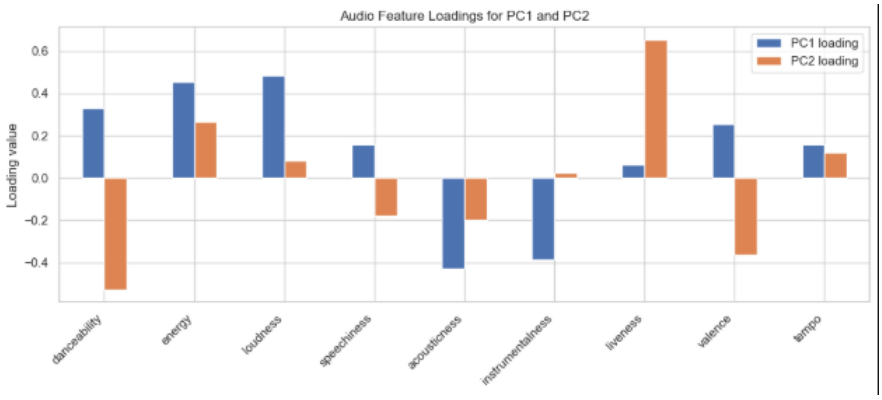
9. PCA Loadings and Visualization

PCA loadings help explain how the original audio features contribute to each principal component. The notebook examined the loadings of PC1 and PC2.

Audio feature	PC1 loading	PC2 loading
Danceability	0.330	−0.526
Energy	0.454	0.266
Loudness	0.483	0.081
Speechiness	0.158	−0.181
Acousticness	−0.428	−0.197
Instrumentalness	−0.386	0.023
Liveness	0.062	0.653
Valence	0.255	−0.364
Tempo	0.158	0.118

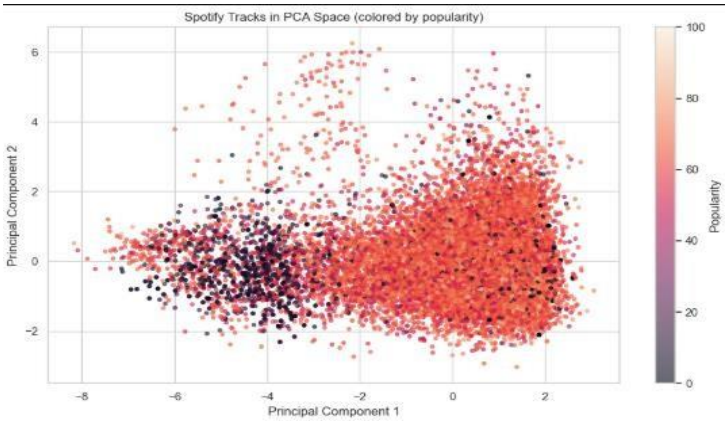
PC1 had stronger positive loadings for loudness and energy, while acousticness and instrumentalness had negative loadings. PC2 had a strong positive loading for liveness and a negative loading for danceability.

These loadings show how the original features contribute to the new dimensions. The sign of a loading indicates direction within the component and should not be interpreted independently as a direct effect on popularity.



PCA loadings chart for PC1 and PC2

A scatter plot of songs in the PC1–PC2 space was also generated, with points coloured by popularity. This visualization helps examine how tracks are distributed across the first two principal components and whether any visible patterns correspond to popularity.



PC1 versus PC2 scatter plot

10. Reduced Dataset Export

After PCA, the transformed data was reduced from nine original audio features to six principal components. Popularity was retained separately for further analysis.

The reduced dataset was exported as **spotify_pca_week7.csv**, containing 19,334 records and seven columns: PC1, PC2, PC3, PC4, PC5, PC6, and popularity. The original cleaned dataset remained unchanged.

This reduced dataset can be used in subsequent analysis and modelling where a smaller feature representation is useful.

11. Team Member Participation

S. No.	Team Member	Roll Number	Contribution
1	Prithesa R	25BAD083	Data cleaning and preparation of audio features
2	Saravanan A	25BAD102	Correlation analysis and interpretation of results
3	Tejas T	25BAD124	PCA implementation and dimensionality reduction
4	Viknesh T	25BAD128	Feature selection, visualization, and report preparation

12. Challenges and Considerations

- The original dataset contained one missing popularity value, so the corresponding record was excluded from the analysis.
- Standardization was necessary because the audio features use different numerical scales.
- Feature selection and PCA serve different purposes: SelectKBest chooses individual original features, while PCA creates new combined features.
- PCA preserves variation rather than guaranteeing that every component is useful for predicting popularity.
- For future predictive modelling, the data should be split into training and testing sets before fitting preprocessing, feature selection, or PCA, to avoid data leakage.

13. Conclusion

Week 7 completed feature selection and dimensionality reduction on the Spotify dataset. Correlation analysis identified the main audio features associated with popularity, and SelectKBest selected loudness, instrumentalness, acousticness, energy, and danceability as the five highest-scoring features. PCA reduced the nine audio features to six principal components while retaining approximately 90.06% of the total variance. The reduced dataset and visualizations provide a useful foundation for further analysis of song popularity.